

Qatar-2b

R-0949

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_180315 · created 2026-07-29T14:19:05+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458192.86735 +/- 0.0006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.066
Transit depth (Rp/Rs)^2	0.00014 +/- 0.015 [%]
Orbital Inclination (inc)	86.99 +/- 1.73
Airmass coefficient 1 (a1)	2.32 +/- 1.62
Airmass coefficient 2 (a2)	-0.67 +/- 0.55
Scatter in the residuals of the lightcurve fit is	60.67 %
Best Comparison Star	#2 - [504, 384]
Optimal Aperture	2.29
Optimal Annulus	3.25
Transit Duration (day)	0.0647 +/- 0.0062

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.001 $\pm$ 0.066 vs 0.16327 (deviation 2.45 σ)
Duration fit vs published	0.0647 d vs 0.0767 d (deviation 1.93 σ)
Mid-time O-C	-0.9 min
Depth SNR	0.0
Residual scatter	60.67 %

Light curve

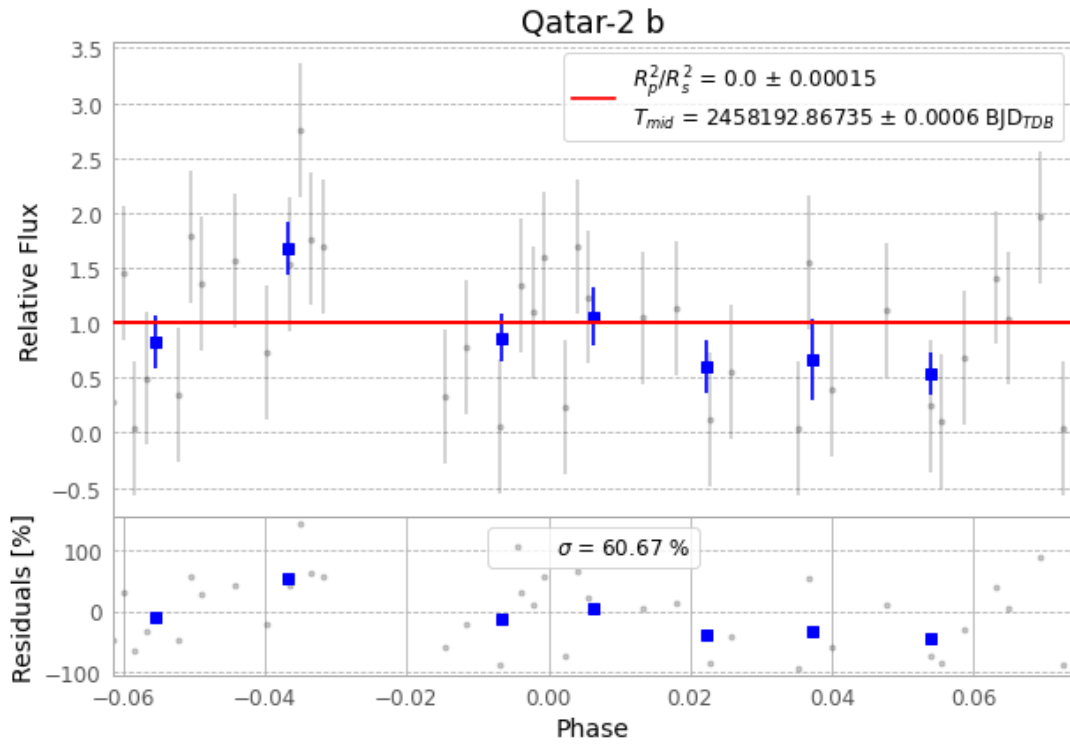


Figure 1 — FinalLightCurve_Qatar-2 b_2018-03-14.png (SHA-256 84f451a5ebbec6fb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [221, 88], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ff27fac9929332f7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

92 FITS frames (60 MB), Qatar-2180315063312.FITS → Qatar-2180315110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-180315.log (d5c1a39f3877...), FinalLightCurve_Qatar-2 b_2018-03-14.png (84f451a5ebbe...), FinalLightCurve_Qatar-2 b_2018-03-14.csv (75763eeb9e4a...)

Compute resources

Wall time 215.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 14:19Z.